Age-Tapered Income Adjustment

A Timing Correction for Lifetime Tax Equity

L McCulloch-James

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Executive Summary

A pound earned at twenty-five differs from a pound earned at sixty. It has a longer compounding horizon, faces tighter liquidity constraints, and can be deployed at a stage of life where the capacity to convert income into realised outcomes is higher. Current tax systems treat income as temporally homogeneous. This introduces systematic mispricing.

This paper sets out an **Age-Tapered Income Adjustment (ATIA)**: a shallow, continuous, symmetric modification of marginal tax rates across the working life, defined over ages 16 to 70. Savings-linked variants are excluded on structural grounds. They favour households already able to defer consumption. The correction is applied directly to income.

The instrument is:

- continuous, with no cliff edges,
- approximately revenue-neutral,
- legally defensible under proportionality,
- implementable within existing PAYE systems.

The Structural Claim

Let income be indexed by age a and earnings level y. The lifetime value of a unit of income has two components: a deferred-value component reflecting the share saved or invested, and a contemporaneous-utility component reflecting the share consumed at the age it is received.

$$V(y,a) \;=\; s(a)\,y\,e^{r(T-a)} \;+\; \big[1-s(a)\big]\,y\,u(a)$$

where $s(a) \in [0,1]$ is the saving rate at age a, r is an effective real return, T is the planning horizon, and $u(a)$ is the health-adjusted marginal utility of contemporaneous consumption.

Both terms decline in a. The deferred term falls as the compounding horizon $T-a$ shortens. The contemporaneous term falls as $u(a)$ falls, since health-adjusted capacity to convert income into realised outcomes declines across the life course, with increasing variance at older ages.

$$\frac{\partial V}{\partial a} < 0$$

The result holds independently of $s(a)$. Earlier income carries higher lifetime value whether it is saved or spent. A tax system that prices income without reference to age treats $V(y,a)$ as constant in a and embeds a persistent distortion.

Why Savings-Based Instruments Fail

Savings-contingent instruments exhibit consistent structural features:

- access requires liquidity,
- benefits accrue to households able to defer consumption,
- relief depends on behaviour rather than position.

Constrained young earners face barriers to access. Liquidity-rich households capture the available relief.

> *Savings instruments redistribute within cohorts. Intertemporal asymmetry remains unchanged.*

Income-side correction applies at the point income arises and reaches all earners without eligibility filters. The 16-year lower bound aligns with the age at which UK income tax liability begins; sub-personal-allowance earners are unaffected by the adjustment in practice.

Within-Cohort Distribution

Proportional scaling of marginal bands delivers a larger absolute reduction to higher-rate payers than to basic-rate payers of the same age. At $\epsilon = 0.10$ and $a = 25$, a higher-rate payer receives a 2.7 percentage point reduction on the higher band; a basic-rate payer receives 1.3 percentage points on the basic band. This is a feature of the rate-scaling design, not the age structure.

The design response is to retain proportional scaling in the default specification, on the grounds that the structural claim above applies at all income levels, and to flag two alternatives for calibration:

- **Capped scaling.** Apply $g(a)$ only up to a defined income threshold (e.g. the higher-rate threshold), leaving income above unaffected.
- **Flat-rate adjustment.** Replace proportional scaling with a fixed percentage-point adjustment to the basic rate only.

The choice between these is a calibration decision, not a structural one, and can be made without altering the instrument's age profile.

Instrument

Let $T'(y)$ denote the existing marginal tax schedule. Define:

$$T'(y,a) = T'(y)\,\big[1 + g(a)\big], \qquad g(a) = \epsilon \cdot \frac{a - a^*}{\Delta}$$

with $a^* = 43$ (midpoint of the 16–70 working range and close to the modal age of peak earnings in UK data), $\Delta = 27$ (half-range), and $\epsilon \in [0.05, 0.10]$.

The adjustment is continuous over age. The schedule introduces no discontinuities.

The default specification scales all marginal bands proportionally and leaves thresholds unchanged. Capped or flat-rate variants, as described above, are available within the same age structure.

Calibration

At $\epsilon = 0.10$ on a 40% base marginal rate:

Age	Adjustment $g(a)$	Effective Rate
16	-10.0%	36.0%
25	-6.7%	37.3%
35	-3.0%	38.8%
43	0.0%	40.0%
55	$+4.4\%$	41.8%
65	$+8.1\%$	43.3%
70	$+10.0\%$	44.0%

Initial deployment can begin at $\epsilon = 0.05$, halving each figure above.

Revenue Position

Let $f(a)$ denote the income-weighted age distribution:

$$\int g(a)\, y(a)\, f(a)\, \mathrm{d}a \approx 0$$

The instrument is approximately revenue-neutral under realistic UK age–income profiles. Residual bias is addressed through calibration of ϵ.

Distributional Effects

Younger earners. Higher net income delivered at the point of earning. Access is universal.

Mid-career. Neutral by construction.

Older earners. Slightly higher effective rates on labour income. The adjustment applies to earned income. Pension-dependent households experience limited exposure.

Asset positions vary across the population and are not assumed in the design.

Precedent

UK fiscal policy includes age-linked structures tied to defined objectives: age-related personal allowances, pension contribution regimes, and age-bounded savings instruments. Policy development trends toward continuous, principled structures. ATIA follows this direction.

Policy development has shifted away from discrete age-banded allowances toward unified and gradually adjusted structures, as seen in the phasing-out of age-related personal allowances and the increasing use of tapers and parameter roll-outs in the tax system [1, 2].

Legal Position

The instrument satisfies established criteria:

- legitimate aim: correction of intertemporal income mispricing,
- proportionality: small, continuous, symmetric adjustments,
- necessity: alternative instruments do not operate at the point of income.

Age differentiation is assessed through justification and proportionality. ATIA aligns with these criteria.

Behavioural Effects

Expected:

- improved early-life financial stability,
- reduced reliance on intergenerational transfers,
- minimal labour supply distortion,
- moderate extension of late-career labour supply.

Continuity of the function limits incentives for income timing strategies.

Implementation Pathway

1. Introduce at $\epsilon = 0.05$ within PAYE systems.
2. Evaluate over one tax cycle.
3. Adjust calibration toward $\epsilon = 0.10$ based on observed outcomes.

The mechanism integrates into existing tax computation structures without additional administrative layers.

Core Claim

A tax system that ignores time embeds a pricing error.

Income carries temporal structure. Policy can reflect that structure directly. Savings-linked relief does not address the underlying distortion. Income-side adjustment operates at the correct point in the system.

A continuous, symmetric taper across ages 16–70:

- aligns taxation with the temporal value of income,
- preserves structural simplicity,
- fits within legal and political constraints.

The proposal reweights taxation across the life course. It aligns tax liability with the timing of income.

References

[1] UK Parliament, House of Commons Library (2019). *Age-related personal allowance.* https://commonslibrary.parliament.uk/research-briefings/sn06158/

[2] Institute for Fiscal Studies (2022). *Reforms, roll-outs and freezes in the tax and benefit system.* https://ifs.org.uk/publications/reforms-roll-outs-and-freezes-tax-and-benefit-system